# Hardware-Aware Quantum Bayesian Learner Compression: Linking Human Belief Updating to NISQ Circuit Complexity

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***Abstract*—We connect quantum models of cognition with the constraints of near-term quantum hardware. A small *quantum Bayesian learner*—a belief state on a two-qubit register, a prior, and a stimulus-dependent evidence channel—learns a similarity-structured categorization task, with the posterior category read out by measurement. Making the learner a hardware-efficient variational circuit, we penalize the real post-transpile two-qubit gate count on IBM's `FakeManila` backend and compress it by greedy structured pruning of individual Heisenberg interaction terms. Across three difficulty levels and five seeds the accuracy–complexity frontier exposes a *capacity boundary*: from two-thirds compression up to the full circuit accuracy is statistically indistinguishable, so structured compression is performance-neutral, yet removing every entangler drops the learner to chance. Learned masks beat random masks at every matched budget (paired Wilcoxon $p \approx 1.7 \times 10^{-8}$), so *which* interactions are kept matters, not only how many. On `ibm_fez` the ordering inverts: the full circuit falls to $0.60$ while the compressed circuit holds $0.83$ ($n$=40, Fisher exact $p$=0.047). On real hardware, capacity that is free in simulation is actively harmful, and hardware-aware compression becomes necessary rather than merely economical.**

***Index Terms*—Quantum cognition, Bayesian learning, variational quantum circuits, hardware-efficient ansatz, NISQ devices, circuit compression.**

## I. Introduction

Bayesian models cast human learning as *belief updating*: a prior over hypotheses revised by evidence [1]; *quantum cognition* represents belief states in Hilbert space as density operators updated by quantum channels [2]. In parallel, NISQ devices [6] have made variational quantum algorithms the dominant near-term paradigm [5], built from hardware-efficient ansätze [3], [4]. Because two-qubit gates dominate the NISQ error budget [6], [7], reducing their number without destroying task performance is a central engineering objective.

We join these threads. The belief is a state on a two-qubit register, the evidence update is a stimulus-conditioned channel, and training is on a similarity-structured categorization task [9]; the update circuit is a Heisenberg-type hardware-efficient ansatz, and a hardware-aware loss penalizes the two-qubit gates remaining after transpilation. The guiding question is concrete: *how much of the circuit can be removed before the learner can no longer represent the task?* We contribute a compact, exactly-simulated quantum Bayesian learner; a hardware-aware cost equal to the real post-transpile two-qubit count, resolved per $R_{XX}/R_{YY}/R_{ZZ}$ term; accuracy–complexity frontiers locating a *capacity boundary*; a matched-budget learned-versus-random ablation; and a single-seed validation on IBM hardware.

## II. Methods

**Belief states and evidence channels.** The belief about a stimulus is a state $\rho$ on $n$=2 qubits. A prior $\rho_0$ is updated by a CPTP channel conditioned on the stimulus $x$, $\mathcal{E}_x(\rho) = \sum_k E_{x,k}\, \rho\, E_{x,k}^\dagger$, with the posterior read out by measurement [2]. A fast *pure-state* model takes $\rho_0 = |0\rangle\langle 0|^{\otimes n}$ with a unitary channel; a literal *mixed-state* model starts from $\mathbb{I}/2^n$ and applies a non-unital channel (a stimulus-dependent Lüders/POVM filter with Bayesian trace renormalization plus amplitude damping). Both are exactly simulated. "Bayesian" is meant operationally: the update is a normalized quantum filter, not a claim about human inference.

**Ansatz and hardware-aware cost.** Each of $D$=4 layers applies single-qubit $R_X, R_Z$ rotations and a Heisenberg entangler $U_{ij} = \exp[-i(\alpha X_i X_j + \beta Y_i Y_j + \gamma Z_i Z_j)]$ realized as $R_{XX}, R_{YY}, R_{ZZ}$ gates, interleaved with data re-uploading [3], [4]. A per-term binary mask $m_{\ell,p} \in \{0,1\}$ selects each interaction term (12 maskable terms; $R_{PP}(0) = \mathbb{I}$ when off) [8]. The cost $N_{2q}(m)$ is the real number of two-qubit gates surviving transpilation to `FakeManila`, SWAPs included, via Qiskit [10]; the full circuit gives $N_{2q}$=24. The objective is $L(\theta, m) = \frac{1}{N}\sum_k \mathrm{BCE}(y_k, \hat{y}_k) + \lambda N_{2q}(m)$. `FakeManila` is a conservative linearly-connected proxy: pruning consumes only the *ranking* it induces over masks, so the deployment device need not be the one the cost is scored against. The readout uses only *single-qubit* Pauli expectations ($\langle Z_0\rangle, \langle Z_1\rangle, \langle X_0\rangle, \langle X_1\rangle$) through a trainable head $\hat{y} = \sigma(w \cdot v + b)$, so joint-feature structure reaches the head only through entanglement — which is why $N_{2q}$ operationally bounds capacity here.

**Compression and task.** Parameters are trained by Adam with exact JAX autodiff (no parameter-shift), verified against Qiskit to $|1 - \text{fidelity}| \approx 4 \times 10^{-16}$. *Greedy structured pruning* [11] repeatedly removes the term whose removal least increases training cross-entropy, warm-starting after each step, tracing the accuracy–$N_{2q}$ frontier from 24 to 0; $\lambda$ then selects an operating point on it. The task is a checkerboard

TABLE I
LEARNED VS. RANDOM MASKS AT MATCHED BUDGET (HARD TASK, MEAN TEST ACCURACY, FIVE SEEDS).

| $N_{2q}$ | 2 | 6 | 10 | 14 | 18 | 22 |
|---|---|---|---|---|---|---|
| Learned | 0.847 | 0.927 | 0.913 | 0.917 | 0.900 | 0.897 |
| Random | 0.742 | 0.814 | 0.794 | 0.826 | 0.809 | 0.836 |
| $\Delta$ | $+.104$ | $+.112$ | $+.119$ | $+.091$ | $+.091$ | $+.061$ |

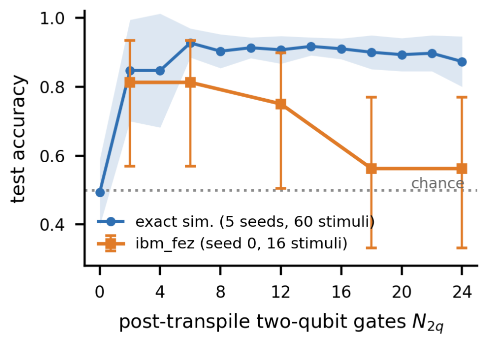


Fig. 1. Exact simulation (5 seeds, 60 stimuli; band $\pm1$ s.d.) versus `ibm_fez` (seed 0, 16 stimuli, 2048 shots; 95% Wilson intervals). The simulated frontier is flat above $N_{2q}$=6; on hardware accuracy declines with $N_{2q}$ and reaches chance for the full circuit.

categorization $y = (\lfloor fx_0 \rfloor + \lfloor fx_1 \rfloor) \bmod 2$ on $[0,1]^2$ — synthetic, inspired by the simplicity principle of [9] rather than taken from it — with difficulty knob $f$=2, 3, 4; 200 balanced stimuli split 70/30 (chance 0.5).

## III. RESULTS

**Capacity boundary.** The learner is well above chance at every nonzero budget (easy/medium $\approx$ 0.94–0.97, hard $\approx$ 0.85–0.93), and from $N_{2q}$=8 to 24 accuracy is flat to within seed-to-seed spread: at one third of the full budget the hard task reaches $0.903 \pm 0.049$ against $0.873 \pm 0.073$ for the full circuit, so two thirds of the entangling structure is redundant. The hard-task optimum lies below full cost (0.927 at $N_{2q}$=6), but the paired advantage is only $+0.053$ and *not* significant at five seeds (paired $t$, $p$=0.12; Wilcoxon $p$=0.13); we therefore claim only that structured compression is performance-neutral, noting the direction reproduces in the density-matrix model. At $N_{2q}$=0 easy and hard fall to chance (0.523, 0.493) while medium retains 0.643 — above chance, but far below its 0.95 plateau. Some entangling structure is thus necessary, and far less than the full circuit supplies.

**Structure beats count.** At matched budgets, greedy masks beat random masks of identical sparsity at *every* budget by $\sim$6–12 points (Table I); per seed the learned mask wins 49 of 55 paired comparisons, mean advantage $+0.083$, one-sided Wilcoxon $p \approx 1.7 \times 10^{-8}$. *Which* entanglers are kept matters, not only how many — and unlike the frontier shape, this result is statistically unambiguous.

**Validation on hardware.** We ran the trained circuits on `ibm_fez`, a 156-qubit Heron device (hard task, seed 0, default mitigation). Across five budgets (16 stimuli, 2048 shots) device accuracy declines monotonically with $N_{2q}$ — 0.81 at $N_{2q}$=2, 6, 0.75 at 12, 0.56 at 18 and 24 — while the five-seed exact frontier stays flat (Fig. 1). At $n$=16 the Wilson intervals are wide and no pair of budgets separates, so that sweep is directional only; the claim rests on a denser endpoint run (40 stimuli, 4096 shots), where the compressed circuit scores 0.825 $[0.68, 0.91]$ against 0.600 $[0.45, 0.74]$ for the full circuit — significant by Fisher's exact test ($p$=0.047) — though in simulation the two are equivalent (0.925 vs 0.950). This is one seed, one device, one calibration: an existence proof, not a characterization. Within that scope, capacity that is free in simulation is actively harmful on the device.

**Robustness.** Under the `FakeManila` noise model (4096 shots, five seeds) the noisy frontier tracks the noiseless one within a few percent, so the savings survive shot noise and gate error. The mixed-state model reproduces every finding (hard 0.883 at $N_{2q}$=6 vs 0.872 at 24; learned>random by $+0.043$; three seeds), and classical baselines on identical splits confirm the difficulty axis.

## IV. CONCLUSION

Greedy structured pruning of a quantum Bayesian learner reveals a three-part capacity boundary. Most entangling structure is redundant — roughly two thirds of the two-qubit gates go at no measurable cost. Some entanglement is nevertheless necessary, since removing all of it drops the easy and hard tasks to chance. And structure beats count: learned masks outperform random masks at every matched budget. On hardware the ordering inverts relative to simulation, so compression is what makes the learner usable at all. Limitations: the device result is a single seed on one backend, $N_{2q}$ is scored on a proxy, and the cognitive framing is an interpretive lens on a controlled circuit experiment, not a claim about human inference. Multi-qubit belief registers and repeated device runs with measured error rates folded into the objective are the natural next steps. Code and logs are public.[1]

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